

# Manfred Diaz

[Website](#) | [Github](#) | [Email](#) | [LinkedIn](#) | [Google Scholar](#)

Montreal, Quebec, Canada

## Education

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|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Sep 2018<br>May 2025 | Ph.D. in Computer Science, University of Montreal, Montreal, Quebec, Canada<br>with Prof. <a href="#">Liam Paull</a> .                                       | <a href="#">Thesis</a> |
| Sep 2016<br>Aug 2018 | M.Sc. in Computer Science, Concordia University, Montreal, Quebec, Canada<br>with Prof. <a href="#">Thomas Fevens</a> and Prof. <a href="#">Liam Paull</a> . | <a href="#">Thesis</a> |
| Sep 2005<br>Jul 2010 | B.Sc. in Computer Science, Universidad de las Ciencias Informáticas, Habana, Cuba<br>with Prof. <a href="#">Eddy Sanchez</a> .                               | Thesis                 |

## Work Experience

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|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| Aug 2025<br><i>Present</i>  | <b>Senior Machine Learning Engineer   DRW Holdings</b><br>in the AI and MASS teams for ML applications to trading.                                                                               | Montreal, CA               |
| June 2025<br><i>Present</i> | <b>Machine Learning and Alignment Theory (MATS) Scholar</b><br>working on safe reward specification and optimization with <a href="#">Michael Dennis</a> .                                       | London, UK<br>Montreal, CA |
| Oct 2023<br>April 2024      | <b>AI Resident   Google X</b><br>Large language models for supply chain understanding, optimization, and control.                                                                                | Mountain<br>View, CA       |
| Jun 2022<br>Sept 2022       | <b>Research Intern   J.P. Morgan AI Research</b><br>with <a href="#">Joshua Lockhart</a> on explainable AI and concept-based explanations.                                                       | London, UK                 |
| May 2021<br>July 2021       | <b>Research Intern   Motional Inc</b><br>with <a href="#">Eric Wolff</a> on motion planning, behaviour prediction, and graph representation learning<br>for autonomous driving.                  | Boston, MA                 |
| May 2019<br>Aug 2019        | <b>Research Intern   Huawei Canada</b><br>with <a href="#">Jun Luo</a> and <a href="#">Peyman Jadmellat</a> on behaviour prediction and graph representation<br>learning for autonomous driving. | Markham, ON                |
| June 2018<br>Aug 2018       | <b>Deep Learning Intern   Mila, University of Montreal</b><br>with Prof. <a href="#">Liam Paull</a> on imitation learning and sim-to-real transfer for autonomous driving.                       | Montreal, QC               |
| Jun 2017<br>Aug 2017        | <b>Software Engineer Intern   Manu3Lab, Anatomy Metrix</b><br>on computer vision, real-time camera calibration, and video streaming.                                                             | Montreal, QC               |
| Sep 2016<br>Jan 2018        | <b>Visiting Researcher   Shared Reality Lab, McGill University</b><br>with Prof. <a href="#">Jeremy Cooperstock</a> on navigation, assistive technologies, and imitation learning.               | Montreal, QC               |
| July 2015<br>Jan 2016       | <b>Lead Software Developer   H3alth Technologies Inc</b><br>on sensor acquisition, point cloud registration, 3D reconstruction.                                                                  | Quito, EC<br>(remote)      |
| Oct 2014<br>Apr 2017        | <b>Lead Software Developer   GoMentr Inc</b><br>on web technologies, single-page applications, Grails Framework.                                                                                 | Quito, EC<br>(remote)      |
| Sep 2010<br>June 2014       | <b>Software Architect   Universidad de las Ciencias Informáticas</b><br>on desktop technologies, .NET Framework, distributed systems.                                                            | Havana, Cuba               |
| Jun 2007<br>Aug 2010        | <b>Software Engineer Intern Co-Op   Universidad de las Ciencias Informáticas</b><br>on desktop technologies, .NET Framework, distributed systems                                                 | Havana, Cuba               |

## Publications

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- 2025 C. Smith, M. Abdulhai, **M. Diaz**, M. Tesic, R. Trivedi, S. Vezhnevets, L. Hammond, J. Clifton, M. Chang, E. A. Duéñez-Guzmán, J. P. Agapiou, J. Matyas, D. Karmon, [Contestants], B. McHale, T. Baarslag, D. Hadfield-Menell, N. Jaques, J. Hernandez-Orallo, J. Z. Leibo. [Evaluating Generalization Capabilities of LLM-Based Agents in Mixed-Motive Scenarios Using Concordia](#). NeurIPS Datasets & Benchmarks Track, 2025 [\[paper\]](#)
- 2025 J. Z. Leibo, A. S. Vezhnevets, W. A. Cunningham, S. Krier, **M. Diaz** and S. Osindero. [Societal and technological progress as sewing an ever-growing, ever-changing, patchy, and polychrome quilt](#). In *arXiv [cs.AI]*. arXiv. [\[preprint\]](#).
- 2025 M. Lanctot, K. Larson, M. Kaisers, Q. Berthet, I. Gemp, **M. Diaz**, R-R. Maura-Rivero, Y. Bachrach, A. Koop, and D. Precup. [Soft Condorcet Optimization for Ranking of General Agents](#). Autonomous Agents and Multi-Agent Systems, 2025 [\[preprint\]](#) [\[paper\]](#) [Best Paper Award](#).
- 2024 J. Z. Leibo, A. S. Vezhnevets, **M. Diaz**, J. P. Agapiou, W. A. Cunningham, P. Sunehag, J. Haas, R. Koster, E. A. Duéñez-Guzmán, W. S. Isaac, G. Piliouras, S. M. Bileschi, I. Rahwan, S. Osindero. [A theory of appropriateness with applications to generative artificial intelligence](#). [\[preprint\]](#)
- 2024 **M. Diaz**, J. Z. Leibo, and L. Paull. [Milnor-Myerson Games and The Principles of Artificial Principal-Agent Problems](#). Finding the Frame: An RLC Workshop for Examining Conceptual Frameworks [\[paper\]](#)
- 2024 **M. Diaz**, L. Paull, and A. Tacchetti. [Rethinking Teacher-Student Curriculum Learning under the Cooperative Mechanics of Experience](#). TMLR. 2024. [\[paper\]](#) [\[preprint\]](#)
- 2022 **M. Diaz**, C. Gauthier, G. Berseth and L. Paull. [Generalization Games for Reinforcement Learning](#). Gamification and Multiagent Solutions Workshop, ICLR 2022. [\[paper\]](#)
- 2021 S. S Gu, **Manfred Diaz**, C D. Freeman, H. Furuta, S. K. Seyed Ghasemipour, A. Raichuk, B David, E. Frey, E. Coumans and O. Bachem. [Braxlines: Fast and Interactive Toolkit for RL-driven Behavior Generation Beyond Reward Maximization](#). [\[paper\]](#) [\[code\]](#)
- 2021 **M. Diaz**, L Paull, and P. S. Castro. [LOCO: Adaptive Exploration in Reinforcement Learning via Local Estimation of Contraction Coefficients](#). Self-supervision for Reinforcement Learning Workshop, ICLR 2021. [\[paper\]](#)
- 2021 **M. Diaz**, T. Fevens, and L. Paull. [Uncertainty-Aware Policy Sampling and Mixing for Safe Interactive Imitation Learning](#). 18th Conference on Robots and Vision 2021 (Oral). [\[paper\]](#) [\[code\]](#)
- 2020 **M. Diaz\***, B. Mehta\* and P. S. Castro. [Bisimulation-Inducing Graph Neural Networks](#). ELLIS Workshop on Geometric and Relational Deep Learning. April 2020. [\[preprint\]](#)
- 2020 B. Mehta, **M. Diaz**, Florian Golemo, Christopher J. Pal, and Liam Paull. [Active Domain Randomization and Safety-Critical Few-Shot Learning](#). [\[preprint\]](#) [\[code\]](#)
- 2019 B. Mehta, **M. Diaz**, F. Golemo, C. J. Pal, and L. Paull. [Active Domain Randomization](#). Proceedings of the Conference on Robot Learning, PMLR 100:1162-1176, 2020. [\[paper\]](#) [\[code\]](#)
- 2018 J. Zilly, J. Tani, B. Considine, B. Mehta, A. F. Daniele, **M. Diaz**, G. Bernasconi, C. Ruch, J. Hakenberg, F. Golemo, A. K. Bowser, M. R. Walter, R. Hristov, S. Mallya, E. Frazzoli, A. Censi, L. Paull. [The AI Driving Olympics at NeurIPS 2018](#). The NeurIPS'18 Competitions, pp 37-68. [\[paper\]](#)
- 2017 **M. Diaz\***, R. Girgis\*, T. Fevens, and J. Cooperstock. [To Veer or Not to Veer: Learning from Experts How to Stay Within the Crosswalk](#). In 2017 IEEE International Conference on Computer Vision Workshop (ICCVW), pp. 1470-1479. IEEE, 2017. (Oral) [\[paper\]](#)
- 2011 **M. Diaz**. [Integración de sistemas de gestión de emergencias con tecnologías móviles](#). In the Ninth Latin American and Caribbean Conference, LACCEI 2011 [\[paper\]](#)

## Workshops, Sessions and Competitions Co-Organized

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|            |                                                                         |                           |
|------------|-------------------------------------------------------------------------|---------------------------|
| 2025       | AAMAS 2025 Tutorial on General Evaluation of AI Agents                  | <a href="#">[website]</a> |
| 2022-      | Multi-Agent Learning Seminar                                            | <a href="#">[website]</a> |
| 2021       | BIG-Gym: A Crowd-Sourcing Challenge for RL Environments and Behaviours  | <a href="#">[website]</a> |
| 2021       | NeurIPS Workshop on Ecological Theory of Reinforcement Learning         | <a href="#">[website]</a> |
| 2018- 2019 | AI-DO: The AI Driving Olympics at NeurIPS 2018, NeurIPS 2019, ICRA 2019 | <a href="#">[website]</a> |

## Honours And Awards

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|-----------|----------------------------------------------------------------------------|
| 2018      | DIRO Excellence Award, University of Montreal                              |
| 2017      | NSERC Create Award, Surgical Innovation, McGill University                 |
| 2016      | Surgical Innovation Fellow, McGill University                              |
| 2010      | Dean List, Summa Cum Laude, Universidad de las Ciencias Informáticas, Cuba |
| 2006      | Bronze Medal, Cuban Inter-College Informatics Contest                      |
| 2003-2004 | Bronze Medal, Cuban National Informatics Contest                           |

## Outreach, Inclusion and Volunteering

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|-----------|-------------------------------------------------------------------------------|
| 2018-     | Reviewer NeurIPS, ICML, ICLR, CoRL, ICRA                                      |
| 2020-2022 | Mentor, LatinX in AI (3 students mentored)                                    |
| 2018-2020 | Duckietown Robotics Outreach with ETH Zurich, TTIC, Brown University and NCTU |

## Talks

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| 2025 | Oxford AI Safety Initiative. The Designer's Paradox. Oxford, UK                                                                                |
| 2025 | MATS Summer Research Talks. The Designer's Paradox. Newspeak House, London, UK                                                                 |
| 2022 | Mila RL Sofa. Generalization Games for Reinforcement Learning. <a href="#">[video]</a>                                                         |
| 2022 | Berkeley Multi-agent Learning Seminar. Generalization Games for Reinforcement Learning.                                                        |
| 2018 | Montreal AI Symposium, Duckietown: A Platform for Teaching Robotics and Machine Learning Research. <a href="#">[video]</a>                     |
| 2017 | Workshop on Assistive Computer Vision and Robotics, ICCV 2017: To Veer or Not to Veer: Learning from Experts How to Stay Within the Crosswalk. |

## Teaching Experience

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| 2018 | Teaching Assistant, IFT 6757 Autonomous Vehicles, University of Montreal       |
| 2017 | Teaching Assistant, SOEN498 Emerging Technologies in HCI, Concordia University |

## Referees

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|----------------|----------------|-------------------------------------|----------------------------------------------------------------------|
| Liam Paull     | Ph. D. Advisor | Professor, University of Montreal   | <a href="mailto:paulll@mila.quebec">paulll@mila.quebec</a>           |
| Joel Z. Leibo  | Collaborator   | Research Scientist, Google DeepMind | <a href="mailto:jlz@deepmind.com">jlz@deepmind.com</a>               |
| Marc Lanctot   | Collaborator   | Research Scientist, Google DeepMind | <a href="mailto:lanctot@deepmind.com">lanctot@deepmind.com</a>       |
| Kate Larson    | Collaborator   | Research Scientist, Google DeepMind | <a href="mailto:katelarson@deepmind.com">katelarson@deepmind.com</a> |
| Michael Dennis | Collaborator   | Research Scientist, Google Deepmind | <a href="mailto:dennismi@deepmind.com">dennismi@deepmind.com</a>     |